



Identification of the presence of crystalline phase in lithiated boron oxide ionic glass conductors

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ABSTRACT

The presence of crystalline phase in low and medium lithium content boron oxide glasses $x\text{Li}_2\text{O}-\text{B}_2\text{O}_3$, with $x = 0.3-2.5$, has been detected by combined infrared and impedance measurements. The borate glass samples apparently consist of one single phase, the vitreous. However, characteristic absorption bands appearing in infrared spectra of samples with $x \geq 0.8$ indicate the presence of crystalline Li_2O . Further evidence is provided by the gradual appearance and disappearance with increasing temperature of an additional electrical relaxation in impedance spectra, corresponding to the crystalline phase present in the borate glasses, which is associated with Li_2O .

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1. Introduction

Vitreous lithium borates are a class of ion-conducting glasses [1,2] that have been considered as potential solid electrolytes in lithium-ion batteries, microbatteries and sensors [3–5]. Boron oxide (B_2O_3) is known to be an excellent network former. The addition of Li_2O into borate glass introduces mobile lithium ions, giving rise to ionic conductivity in the resulting $x\text{Li}_2\text{O}-\text{B}_2\text{O}_3$ solid solution [1,2]. In principle, higher lithium content implies greater ionic conductivity. However, the presence of crystalline phase in high lithium content borate glasses, distributed in the form of islands within the vitreous phase, hinders further increase in ionic conductivity with increasing lithium content [6]. We have recently presented a study regarding the influence of the two-phase structure of high lithium content borates on the ionic conductivity [7]. The presence of two material phases, obvious from direct optical inspection, was reflected in impedance measurements by the appearance of two semicircular arcs in complex plane plots, as well as in infrared (IR) spectra by additional characteristic absorption peaks [6–8]. In this paper, we extend our method of study to medium and, mainly, to low lithium content $x\text{Li}_2\text{O}-\text{B}_2\text{O}_3$ borate

glasses, in an attempt to provide evidence as to the presence of the crystalline phase in such cases, where scanning electron microscopy (SEM) fails to do so. The structural and electrical properties of the lithium borates were studied by IR and impedance techniques, and the critical lithium content, above which the crystalline phase is formed within the vitreous phase, was identified.

2. Experimental

The $x\text{Li}_2\text{O}-\text{B}_2\text{O}_3$ glass samples, with x ranging from 0.3 to 2.5, were prepared by melting Li_2CO_3 and B_2O_3 precursors, and shaped in the form of thin sheets by quenching. As Li_2CO_3 is decomposed to Li_2O by heating, giving off CO_2 gas, carbon is removed from the final product. Details have been reported elsewhere [6]. All prepared samples were completely transparent, an indication that the as formed borate glasses apparently consist of one single phase only, the vitreous, as is demonstrated in Fig. 1 by the typical SEM picture of the $x = 1.5$ sample. Nevertheless, further investigation of the samples showed that a second phase, the crystalline, is also present. Identification of the crystalline phase was performed by two experimental methods, optical and electrical [8]. IR spectra of the samples, encapsulated in KBr, were recorded at 2 cm^{-1} resolution by a Fourier transform infrared (FTIR) spectrometer. The instrument background was applied as reference for the absorbance spectra. In order to obtain electrical measurements,

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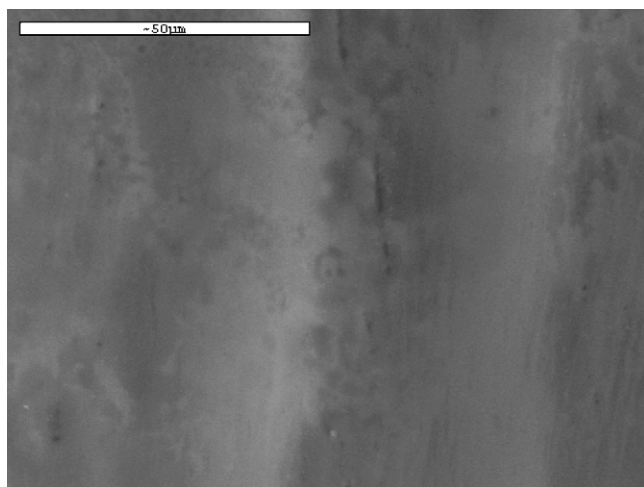


Fig. 1. SEM picture of the $x = 1.5$ sample.

gold electrodes were formed on both sides of the samples by thermal evaporation. Impedance spectra were recorded, as a function of temperature, in the frequency range from 1 MHz down to 10 mHz. The temperature range covered was from 57 to 312 °C [7,8]. During measurement, the samples were kept in vacuum of the order of 1 Pa.

3. Results and discussion

The IR absorption spectra of all samples, corresponding to x ranging from 0.3 to 2.5, are shown in Fig. 2. For $x \geq 0.8$, the spectra feature broad noisy peaks located at the low-frequency end of the IR region, beyond 600 cm^{-1} . Peak intensity definitely increases with increasing lithium content. Pure Li_2O is known to exhibit a broad absorption band centered at 450–500 cm^{-1} [9]. Therefore, these peaks can be attributed to Li_2O being present in the samples with $x \geq 0.8$. The spectrum of the plain B_2O_3 glass material is given in Ref. [1,2], pp. 245 and 150, respectively, for comparison. In the case of high lithium content borate glasses that were previously investigated, it was concluded that the discrete islands of the crystalline phase are mainly associated with Li_2O [8]. In consequence, the appearance of the Li_2O absorption band in the spectra corresponding to $x \geq 0.8$ is a direct indication of the formation of the crystalline phase in these samples. The characteristic peak due to Li_2O is absent from the spectra of samples corresponding to $x = 0.3$

and 0.5, and it may therefore be stated that these samples consist of only the vitreous phase.

Other peaks in the IR spectra are attributed to B–O bond vibrations of elementary atomic units of the various borate anionic groups making up the glass structure. The structure of low lithium content borate glasses is well known in the literature [1,2,10–13], and has been described in detail in previous reports [6,8]. Besides the absence of the Li_2O peak, the spectra of samples $x = 0.3$ and 0.5 differ from all the rest in another respect. Different absorption bands, in relation to their position in the frequency spectrum as well as their microscopic origin, are observed within the spectral region 1100–1500 cm^{-1} . It is generally accepted that peaks appearing in the region 1200–1450 cm^{-1} are attributed to B–O bond stretching vibrations of triangular $[\text{BO}_3]$ elementary units. The analogous vibrations of tetrahedral $[\text{BO}_4]$ units are associated with peaks that appear in the region 850–1100 cm^{-1} . The peak located at about 700 cm^{-1} is attributed to B–O–B bond bending vibrations of bridging oxygen atoms [1,2]. With increasing lithium content, this peak is slightly shifted from 700 cm^{-1} for $x = 0.3$ to 735 cm^{-1} for $x = 2.5$.

In pure vitreous B_2O_3 , the absorption band due to $[\text{BO}_3]$ units, located at 1200–1450 cm^{-1} , exhibits two sharp peaks at 1250 and 1380 cm^{-1} . The peak at 1250 cm^{-1} is associated with characteristic vibration modes of boroxol rings [10]. The addition of increasing Li_2O content in the borate glass results in gradual depolymerization of the covalent structural network and the disappearance of boroxol rings [1,2]. Indeed, in Fig. 2 the 1250 cm^{-1} peak is present for the $x = 0.3$ sample, whereas for $x = 0.5$ it has already been eliminated. The peak at 1380 cm^{-1} is also observed in the spectra of lithiated borate glasses as a broadened absorption band, which is shifted toward larger wavenumbers with increasing lithium content. In Fig. 2, this particular band is centered at 1380 cm^{-1} for $x = 0.3$ and 0.5, and at 1440 cm^{-1} for $x = 0.8$ –2.5. In the spectra of samples with $x \geq 0.8$, a new absorption band appears at 1220 cm^{-1} , which is attributed to B–O $^-$ bond stretching vibrations of $[\text{BO}_3]$ units containing non-bridging oxygen atoms. The peaks at 1220 and 1440 cm^{-1} are indicative of the presence of the orthoborate BO_3^{3-} and pyroborate $\text{B}_2\text{O}_5^{4-}$ anions in the samples with $x \geq 0.8$ [1,2,13]. The broad absorption band in the region 850–1100 cm^{-1} is not exhibited by pure vitreous B_2O_3 , and is indicative of the presence of the triborate $[\text{B}_3\text{O}_5]^-$, tetraborate $[\text{B}_4\text{O}_{13}]^{2-}$, pentaborate $[\text{B}_5\text{O}_{11}]^-$ and diborate $[\text{B}_2\text{O}_7]^{2-}$ groups in all of the samples [1,2,10].

The obvious differentiation of the samples into two groups, which can be seen in the optical measurements of Fig. 2, is also consistently followed in electrical measurements. This differentiation is associated with the formation of the crystalline phase in the borate glasses corresponding to $x \geq 0.8$, as was stated earlier. Likewise, impedance spectra measured from the samples with $x \geq 0.8$ exhibit two electrical relaxation time constants, an experimental fact which is characteristic of two-phase systems [7,14]. On the contrary, the samples $x = 0.3$ and 0.5 exhibit only one time constant, indicating the single-phase composition of these samples.

The Nyquist plot of an impedance spectrum obtained from the $x = 0.5$ sample is shown in Fig. 3. The single semicircular arc, corresponding to the relaxation time constant, along with the low-frequency electrode polarization line, is observed in the plot in the whole temperature range covered during the measurements. The ionic conductivity Arrhenius plot for sample $x = 0.5$ is shown in Fig. 4. The equivalent circuit model used for impedance spectra analysis is contained as an insert schematic [7,14]. The arc in the Nyquist plot is reproduced by the parallel combination of constant phase element (CPE) Q_1 and resistor R_1 . The other CPE Q_2 in the circuit is used to model electrode polarization effects. To account for sample dimensions, the value of R_1 , returned by data fitting, is converted to yield the ionic conductivity σ_1 of the vitreous single phase constituting the sample. The results of activation energy E_V and resistivity ρ_V at room temperature of the vitreous phase,

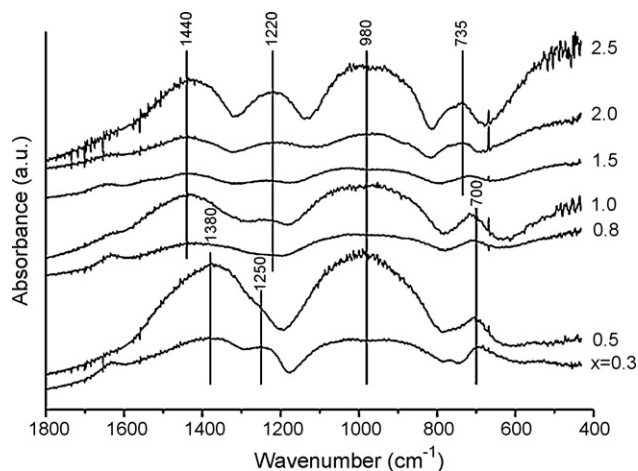


Fig. 2. IR absorption spectra of the low and medium lithium content borate samples.

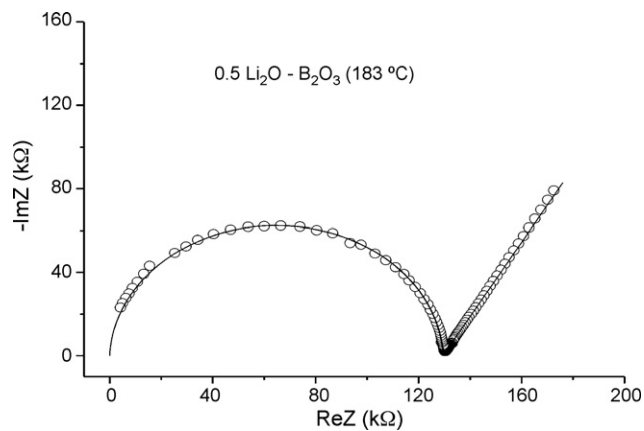


Fig. 3. Nyquist plot of the impedance spectrum measured from the $x=0.5$ sample at 183 °C. The presence of only one semicircular arc is indicative of the single-phase composition of the sample.

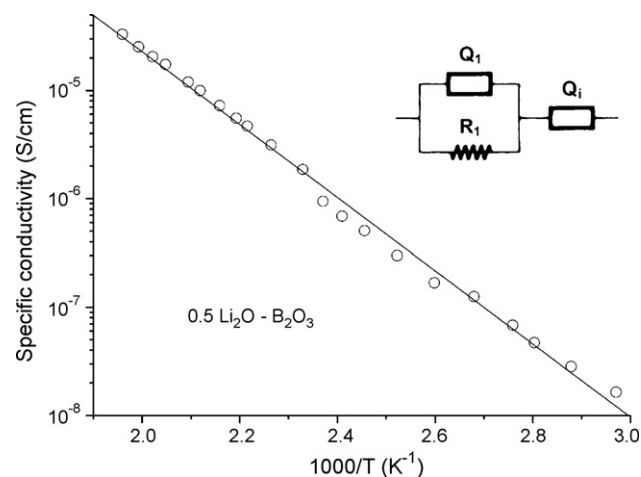


Fig. 4. Arrhenius plot of the $x=0.5$ sample. In the insert schematic, the equivalent circuit used for impedance spectra fitting.

extracted from the Arrhenius plots of all samples, are contained in Table 1.

Figs. 5 and 6 show the Nyquist plots of two impedance spectra, obtained from the $x=0.8$ sample at different temperatures. The two relaxation time constants, typically occurring in two-phase systems, are revealed from the presence of two consecutive semicircular arcs in the plots, followed by the low-frequency polarization line. Each arc is associated with one of the constituent phases [7,14]. However, it turns out that the second arc is conspicuously present only in a small temperature range, extending in any case to no more than 50 °C. With increasing temperature, this middle-frequency second arc continuously becomes smaller in comparison to the high-frequency first arc, until it finally disappears.

Table 1

Calculated ionic conductivity activation energies E_V , E_C of the vitreous and crystalline phases, respectively, and resistivity ρ_V of the vitreous phase at room temperature for all the measured lithium borate samples.

x	E_V (eV)	E_C (eV)	ρ_V (Ω cm) at 25 °C
0.3	0.828 ± 0.011	–	8.35×10^{11}
0.5	0.666 ± 0.011	–	1.62×10^9
0.8	0.641 ± 0.002	1.805 ± 0.083	5.69×10^8
1.5	0.525 ± 0.016	2.419 ± 0.186	6.22×10^7

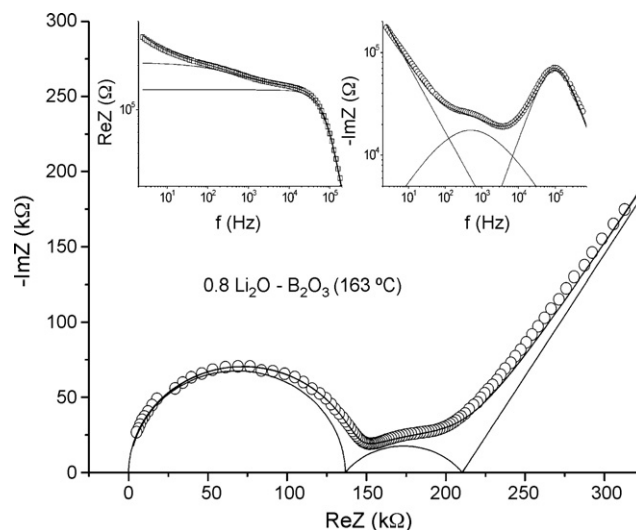


Fig. 5. Nyquist plot of the impedance spectrum measured from the $x=0.8$ sample at 163 °C. In the insert, Bode plots of the real and imaginary parts of the impedance. The second semicircular arc appears with increasing temperature, revealing the presence of the crystalline phase.

This effect, illustrated in Figs. 5 and 6, is due to overlap in the impedance plot of the second arc with either the low-frequency electrode polarization tail at low temperatures, or the first arc at high temperatures. At adequately low temperatures, this overlap causes the second arc to be completely concealed from the plot, as if being non-existent. In the time domain, overlapping implies that the time scale required for the ionic conduction process in the corresponding second phase to take place is much larger than the time scale in which the electrodes are polarized and therefore hinder the flow of ionic charge carriers. As a result, only the conduction process taking place in the primary phase is revealed in the impedance plot, as if the glass sample is comprised of one single phase, the vitreous. On the other hand, at relatively high temperatures, the ionic conduction process in the second phase becomes faster than in the primary phase. In effect, the kinetics of the overall conduction process is determined by the slower of the two consecutive processes taking place in different phases. Consequently, the properties of only the primary sample phase, the vitreous, are again depicted in the impedance plot. To our knowledge, the above remarkable

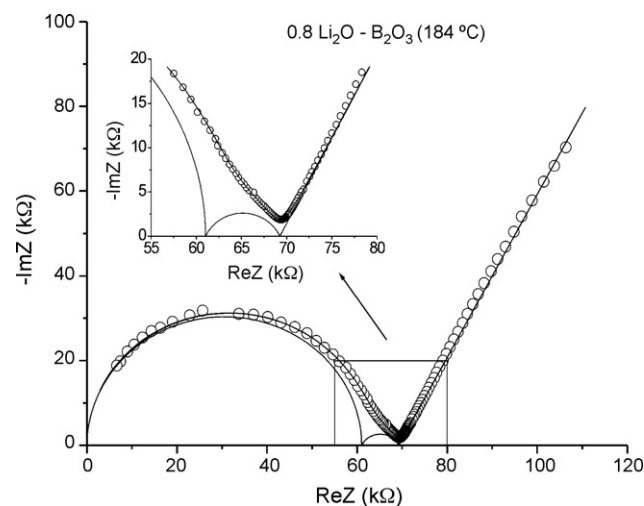


Fig. 6. Nyquist plot of the impedance spectrum measured from the $x=0.8$ sample at 184 °C. In the insert, plot detail zooming in the region of the second semicircular arc. The second arc subsequently disappears with further increasing temperature.

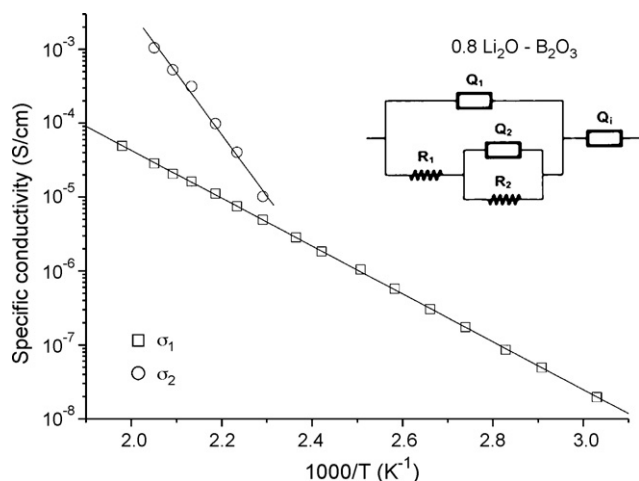


Fig. 7. Arrhenius plot of the $x=0.8$ sample, corresponding to the vitreous (σ_1) and crystalline (σ_2) phases. In the insert schematic, the equivalent circuit model used for impedance spectra fitting.

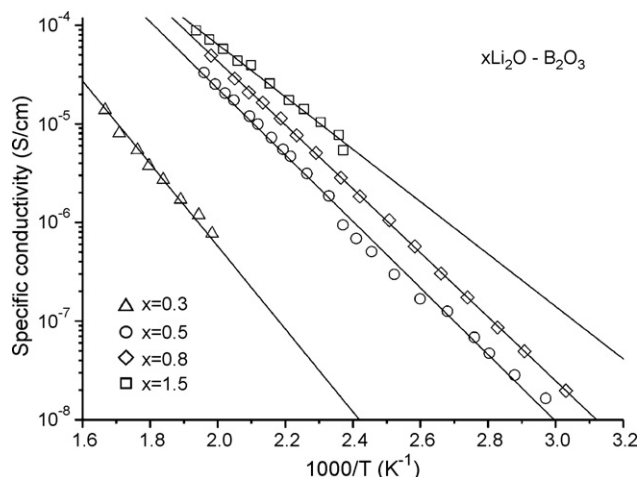


Fig. 8. Arrhenius plots of the vitreous phase ionic conductivity for all the measured samples.

behavior of the middle-frequency arc, appearing and subsequently disappearing with increasing temperature in impedance plots, has not been reported in the literature.

The Arrhenius plot for sample $x=0.8$ is shown in Fig. 7. In order to describe the electrical properties of two-phase systems, we have proposed an equivalent circuit model [7], which reduces to the impedance fitting circuit shown in the insert schematic of Fig. 7. The two semicircular arcs are reproduced by including the equal in number couples of CPEs and resistors (Q_1, R_1) and (Q_2, R_2). The CPE Q_i is again used to account for electrode polarization effects [7,14]. The values of R_1 and R_2 , returned by impedance spectra fitting, are converted accordingly, considering sample dimensions, to yield the macroscopic conductivities σ_1 and σ_2 of the two phases, vitreous and crystalline. The activation energies of ionic conduction in the two phases, E_V and E_C , respectively, are calculated from the slope of the Arrhenius plots of the corresponding conductivities σ_1 and σ_2 [7,8], and are listed in Table 1, where phase parameters obtained from all the measured samples have been collected.

The Arrhenius plots of the vitreous phase ionic conductivity σ_1 for all the measured samples are shown in Fig. 8. From Table 1 and Fig. 8, it is deduced that with increasing lithium content the resis-

tivity ρ_V and activation energy E_V of borate glasses continuously decrease. As for the activation energy E_C of the crystalline phase, wherever present, the results contained in Table 1 are similar to those turned out in our earlier investigation of the analogous high lithium content borate glasses, in which the presence of the crystalline phase is unmistakable [7,8]. This similarity provides further evidence as to the identification of the crystalline phase being the origin of the second relaxation that is observed in impedance spectra, in the case of low and medium lithium content borate glasses studied in the present report as well. Because of the unknown, and most probably complex, geometry characterizing the microscopic distribution of the crystalline phase within the glass samples [6], it is impossible to derive credible results for the resistivity of the pure crystalline phase from the values of R_2 returned by fitting [7], and for this reason such results are missing from Table 1. On the contrary, the analogous results regarding the vitreous phase are in absolute agreement with those already known in the literature [1,2], a fact which adds confidence to the findings of our investigation.

4. Conclusions

Low and medium lithium content boron oxide solid electrolytes $x\text{Li}_2\text{O}-\text{B}_2\text{O}_3$, with $x=0.3-2.5$, prepared by quenching, appear to be glassy one-phase systems. Nevertheless, detailed investigation turns out that this is true only for $x<0.8$. For $x\geq 0.8$ a second, crystalline phase is present, which is attributed to Li_2O on the basis of IR measurements. The crystalline phase is also identified by impedance spectroscopy measurements, but only in a limited temperature range, less than 50°C . The additional semicircular arc in impedance Nyquist plots, revealing the presence of the crystalline phase, appears and subsequently disappears with increasing temperature. By employing equivalent circuit fitting, the electrical properties of the two structural phases, vitreous and crystalline, were evaluated.

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